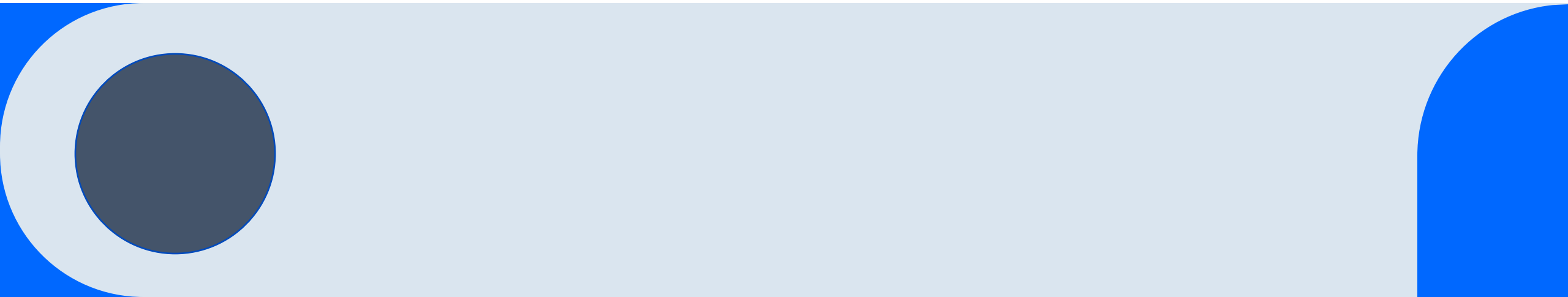
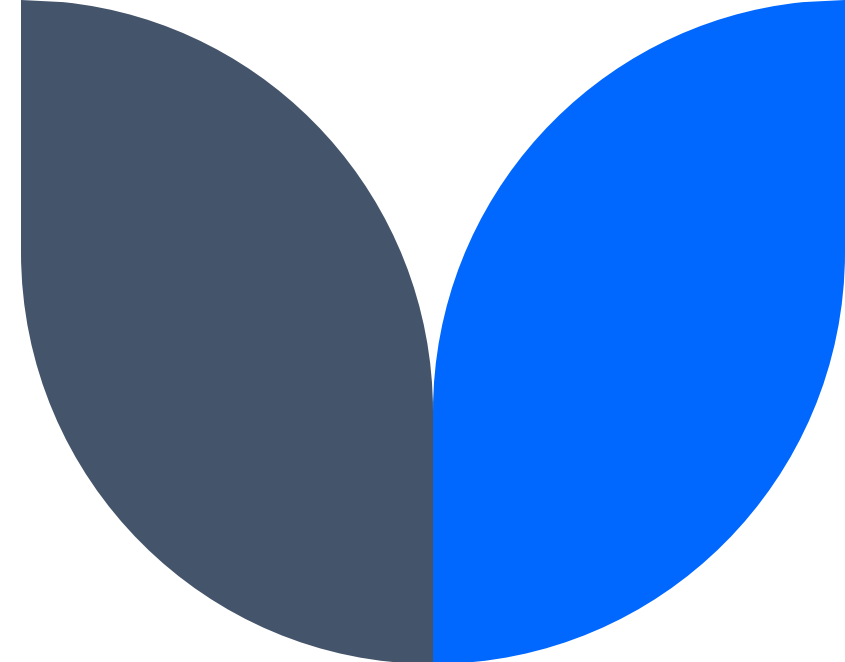




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February 14 2025
Module 7.2 - Assignment:
Pager Rotation Duties

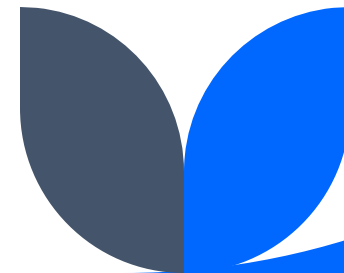


What is Pager Rotation?

- In DevOps, **pager rotation** refers to an on-call system where team members take turns being responsible for handling urgent production issues.
- Ensures **continuous system reliability** and minimizes downtime.

Why is Pager Rotation important?

- Supports incident management.
- Ensures **high availability** of services.
- Distributes workload fairly among team members.



Industry Best Practices

Define Clear On-Call Schedules

- Use tools like **PagerDuty**, **OpsGenie**, or **VictorOps**.
- Rotate shifts fairly to avoid burnout.

Set Reasonable Alert Thresholds

- Avoid false positives (reduce alert fatigue).
- Use **SLOs (Service Level Objectives)** to define acceptable error budgets.

Ensure Proper Handoffs

- Conduct **shift handover meetings**.
- Use documentation (e.g., incident logs, playbooks).



Incident Response and Resolution

Triage Process

- Classify alerts (Critical, High, Medium, Low).
- Use **runbooks** for predefined responses.

Escalation Policy

- Define **who gets paged next** if an issue is unresolved.
- Escalate within **a set timeframe** (e.g., 15-30 mins without response).



Tools for Pager Rotation

Monitoring & Alerting:

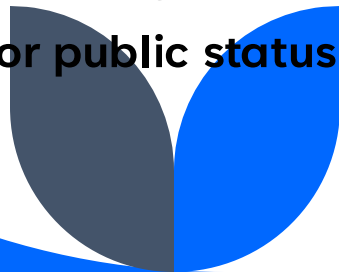
- Prometheus + Alertmanager
- New Relic
- Datadog

On-Call Scheduling & Escalations:

- PagerDuty
- OpsGenie
- VictorOps

Incident Communication:

- Slack / MS Teams integrations
- StatusPage.io for public status updates



Avoiding Burnout

- **Limit On-Call Shifts Per Engineer**
- No more than **one week per month**.
- If employee is full time (40 hr a week) ensure staffing is up to par to avoid overtime.

- **Ensure Adequate Rest Periods**
- No back-to-back on-call rotations.
- Do not force overtime on staff (occasional is understood, excessive leads to unsavory results due to exhaustion).

- **Compensate On-Call Work**
- Provide **monetary incentives** or **extra PTO**.
- Overtime pay must be given when met

Measuring Effectiveness

Key Metrics to Track:

- **MTTA (Mean Time to Acknowledge)**
- **MTTR (Mean Time to Resolve)**
- **Number of After-Hours Pages**
- **Percentage of False Alarms**

Continuous Improvement

- Hold **post-incident retrospectives**.
- Improve **runbooks & automation**.

Case Study - Google SRE Practices

- Google **Site Reliability Engineering (SRE)** model follows strict pager rotation policies.
- Engineers **spend no more than 50%** of their time on-call duties.
- Use **error budgets** to balance reliability and innovation.
- Source: **Google SRE Book** (Beyer et al., 2016).



Sources:

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- Nichols C, 2025, Beyer, B., Jones, C., Petoff, J., & Murphy, N. (2016). *Site Reliability Engineering: How Google Runs Production Systems*. O'Reilly Media.
- PagerDuty. (2023). *The Ultimate On-Call Guide: Best Practices for Incident Response*. <https://www.pagerduty.com>
- Google Cloud. (2023). *SRE Principles*. <https://sre.google>